AMRITSAR GROUP OF COLLEGES

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Project Report

On

LIBRARY MANAGEMENT SYSTEM

Submitted in the partial fulfilment of the requirement for the award of a degree of

Bachelor of Technology

In

COMPUTER SCIENCE AND ENGINEERING

Batch (2023-2027)

Submitted to

Department of CSE

Submitted by

Putul Kumari (2411926)

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

ABSTRACT

The Library Management System is a web-based application developed using the MERN stack (MongoDB, Express.js, React.js, Node.js), aimed at automating the day-to-day operations of a library. The system provides a user-friendly interface for both librarians and members, allowing efficient management of books, members, book issuance, returns, and fines.

Using MongoDB as the database, all records of books, users, and transactions are securely stored and easily retrievable. Express.js and Node.js handle the backend operations, including authentication, book management, and transaction processing, while React.js provides a dynamic and responsive frontend for smooth user interaction.

The system also integrates features like advanced search, real-time availability updates, and report generation, reducing manual effort and errors. Test cases have been implemented to ensure functionality, reliability, and performance. The application is scalable and can be extended with future enhancements like mobile access, online book reservations, and digital content integration.

Overall, the Library Management System using MERN technology offers a modern, efficient, and reliable solution for managing library operations in educational institutions and public libraries.

ACKNOWLEDGEMENT

We express our deep gratitude to our mentor for continuous guidance, valuable feedback, and encouragement during the completion of this training project. We also extend our sincere thanks to Dr. Sandeep Kad, Head of the Department, and all faculty members of the Computer Science & Engineering Department, Amritsar Group of Colleges, for their cooperation and support. This project has helped us gain valuable insights into the practical aspects of Software Engineering and enhanced our understanding of how theoretical knowledge can be applied to real-world applications. It has been a truly enriching learning experience for us.

DECLARATION

We hereby declare that the project work entitled 'Library Management System' is an authentic record of our own work carried out, in partial fulfilment of the requirement for the award of the degree of Bachelor of Technology (CSE) from Amritsar Group of Colleges, Amritsar.

Putul Kumari (2411926)

SOFTWARE REQUIREMENT SPECIFICATION (SRS)

PROJECT TITLE: LIBRARY MANAGEMENT SYSTEM

1. INTRODUCTION

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to provide a detailed description of the Library Management System (LMS). It defines the system's objectives, functions, constraints, and performance requirements. This document serves as a reference for developers, designers, testers, and stakeholders to ensure the system is developed according to expected features and functionalities.

1.2 Scope

The Library Management System aims to automate and simplify library operations such as book management, member registration, book issuing, returning, fine calculation, and searching for books. The system supports both librarians and users to access services efficiently. It will reduce manual work, improve record accuracy, and provide quick information retrieval. The system covers:

- Maintaining records of books and members
- Issuing and returning books
- Managing fines for late submissions
- Searching books by various filters
- Generating reports for library administration

1.3 Objectives

The main objectives of the Library Management System are:

- To automate day-to-day library operations for efficiency
- To maintain accurate and secure records of books and members
- To enable quick book search and availability status
- To reduce paperwork and manual errors
- To manage issue/return transactions and fine calculations easily
- To provide a user-friendly interface for librarians and members

2. OVERALL DESCRIPTION

2.1 Product Perspective

The Library Management System is an independent software application designed to replace the existing manual library process. It provides automated functionalities for book management, member management, and transaction tracking. The system will store all data in a centralized database, enabling secure and fast retrieval of information. It will be used within the library premises by authorized users such as librarians and registered members.

2.2 Product Functions

Key functions of the Library Management System include:

- Managing book details: add, update, delete, and view book records
- Member registration and management
- Issuing and returning books
- Searching for books by title, author, category, or availability
- Automatic fine calculation for overdue books
- Providing reports on issued books, returned books, and defaulters
- Maintaining secure access with login authentication for librarian and member roles

2.3 User Classes and Characteristics

The system distinguishes between different user roles:

User Class	Characteristics	Access Level
Librarian/Admin	Basic computer knowledge; manages all library operations	Full access to book, member, and report management
Library Member/Student	Basic understanding of system; primarily borrowers	Limited access to search books and view personal transactions

Both user classes require login credentials to access the system.

2.4 Operating Environment

The Library Management System will operate under the following environment:

- **Operating System:** Windows / Linux / macOS
- **Backend:** MySQL / SQL Server or any relational database
- **Frontend:** Web-based interface (HTML, CSS, JavaScript) / Desktop GUI

- **Hardware Requirements:** Standard computer system with minimum 4GB RAM and 20GB storage
- **Network:** Local network or Internet connection (if cloud-based)

2.5 Design and Implementation Constraints

- The system must comply with database normalization rules to ensure data integrity.
- Only authorized users should be able to access and manipulate data.
- System must handle concurrent users without performance issues.
- All data should be backed up regularly to prevent loss.
- The interface design should follow usability standards for accessibility.
- Compatibility must be ensured with existing hardware and software infrastructure.

3. FUNCTIONAL REQUIREMENTS

3.1 User Authentication Requirements

- The system shall allow users to register and create an account.
- The system shall allow users (librarian and members) to log in using valid credentials.
- The system shall validate login details and restrict unauthorized access.
- The system shall allow users to log out securely.

3.2 Book Management Requirements (Librarian Only)

- The system shall allow librarians to add new books with details (title, author, ISBN, category, publisher, quantity).
- The system shall allow librarians to update existing book information.
- The system shall allow librarians to delete a book record if required.
- The system shall maintain stock availability of books.

3.3 Member Management Requirements (Librarian Only)

- The system shall allow librarians to add and update member information.
- The system shall maintain unique identification for each member.
- The system shall allow librarians to deactivate membership if needed.

3.4 Book Issue & Return Requirements

- The system shall allow librarians to issue books to registered members.
- The system shall record the issue date and due date automatically.
- The system shall allow librarians to return books and update their status.
- The system shall prevent issuing books that are out of stock.
- The system shall calculate fines automatically for overdue returns.

3.5 Book Search and View Requirements

- The system shall allow users to search books by title, author, category, or ISBN.
- The system shall show book availability status (Issued / Available).
- The system shall allow users to view detailed book information.

3.6 Report Generation Requirements (Admin/Librarian Only)

- The system shall generate reports of:
    - List of issued and returned books
    - Member borrowing history
    - Overdue books and fines collected
    - Total books in library and category-wise details

3.7 Notification Requirements

- The system shall notify librarians about overdue books.
- The system shall alert members about due dates or fines (if notification system integrated).

3.8 Security Requirements

- The system shall restrict data manipulation to authorized users only.
- The system shall maintain audit logs of issued and returned books.

4. NON-FUNCTIONAL REQUIREMENTS

4.1 Performance Requirements

- The system shall respond to user actions (search, login, book issue/return) within 3 seconds.
- The system shall support multiple users accessing the system concurrently without performance degradation.
- The database retrieval and storage functions must be optimized to handle large amounts of records.

4.2 Reliability Requirements

- The system shall ensure that all data operations are executed accurately without data loss.
- The system shall be available for use **99% of the time** during working hours.
- System failures or crashes should not affect stored information.

4.3 Security Requirements

- Login authentication shall be implemented to secure user access.
- Passwords shall be stored in encrypted form to protect user credentials.
- Only authorized roles (librarian/admin) can access and modify sensitive records.
- The system must protect the database from unauthorized read/write access.

4.4 Usability Requirements

- The system shall provide a user-friendly interface for easy interaction.
- Users should be able to perform major tasks with minimal training.
- All functionality shall be clearly labeled and easy to navigate.

4.5 Maintainability Requirements

- The system code shall be modular and easy to update or modify.
- System documentation shall be provided for future upgrades or maintenance.
- Errors and logs must be recorded so issues can be identified quickly.

4.6 Portability Requirements

- The system shall be compatible with multiple operating systems (Windows, Linux, macOS).
- The system shall run on standard hardware configurations.

- The software should be easily installable on different machines without special setup requirements.

4.7 Backup & Recovery Requirements

- Regular automatic backup of database must be performed to prevent data loss.
- The system shall support immediate restoration of data in case of failure.

5. FUTURE ENHANCEMENTS

While the current Library Management System is designed to efficiently handle day-to-day library operations, there are several potential improvements that can be implemented in the future to enhance functionality, usability, and scalability:

5.1 Online Access for Members

- Enable library members to search and reserve books online through a web or mobile portal.
- Allow members to renew books and check due dates remotely.

5.2 Integration with Digital Resources

- Incorporate e-books, journals, and digital content into the system.
- Allow members to read or download e-resources directly through the system.

5.3 Automated Notification System

- Implement SMS or email notifications for overdue books, fines, or new arrivals.
- Automatic reminders for book return dates to improve timely returns.

5.4 Advanced Analytics and Reporting

- Provide detailed reports and analytics, such as popular books, member borrowing trends, and inventory statistics.
- Enable predictive insights to optimize book procurement and library management.

5.5 Mobile Application

- Develop a mobile application for Android and iOS for easier access.

- Allow users to search, reserve, and manage their accounts on the go.

5.6 Cloud-Based System

- Move the system to a cloud platform for remote accessibility and improved scalability.
- Ensure data is synchronized in real-time for multiple branches of a library.

5.7 Multi-Language Support

- Provide support for multiple languages to make the system accessible to a broader user base.

5.8 Integration with RFID/Barcode Scanning

- Implement automated book issuing and returning using RFID tags or barcodes.
- Reduce manual effort and increase speed of transactions.

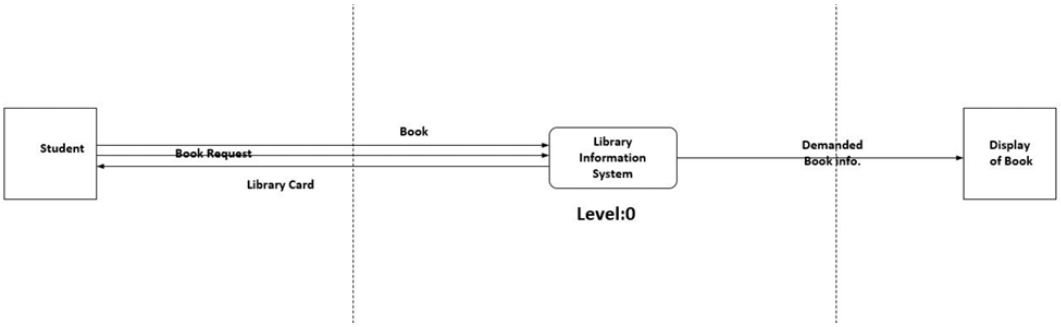
6. CONCLUSION

The Library Management System is designed to streamline and automate the traditional library operations, making them more efficient, accurate, and user-friendly. By implementing this system, libraries can effectively manage book inventories, member records, book issuance and returns, fines, and reporting, reducing manual work and errors.

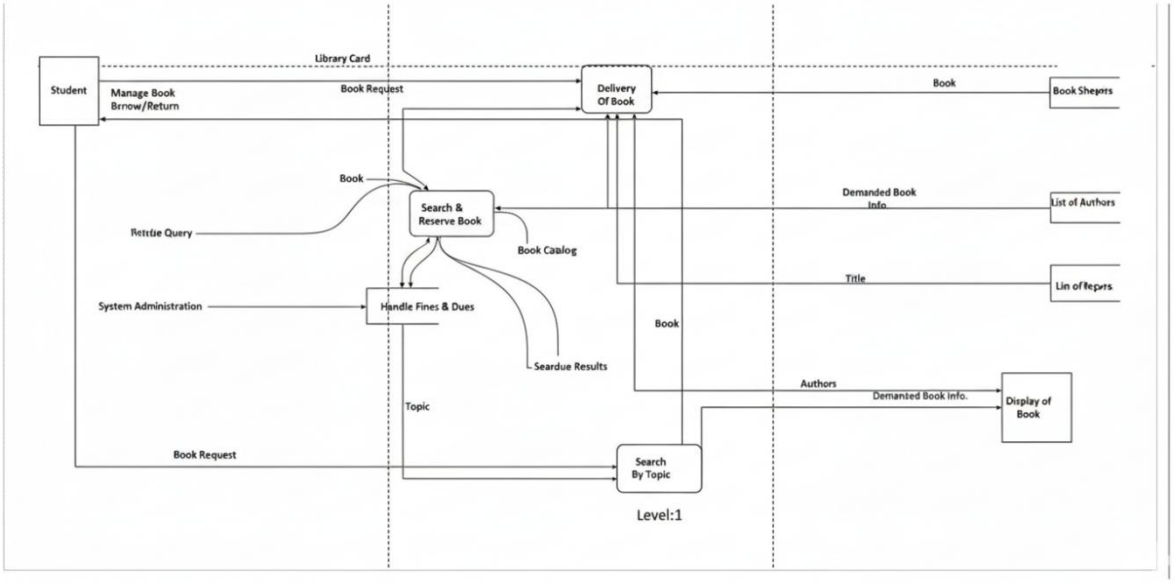
The system not only improves operational efficiency but also enhances the user experience for both librarians and members. Future enhancements such as online access, mobile integration, and advanced analytics can further extend the system's capabilities, making it a comprehensive and scalable solution for modern libraries.

This SRS document provides a clear blueprint for developers, stakeholders, and users, ensuring that the system meets its functional and non-functional requirements while remaining secure, reliable, and maintainable.

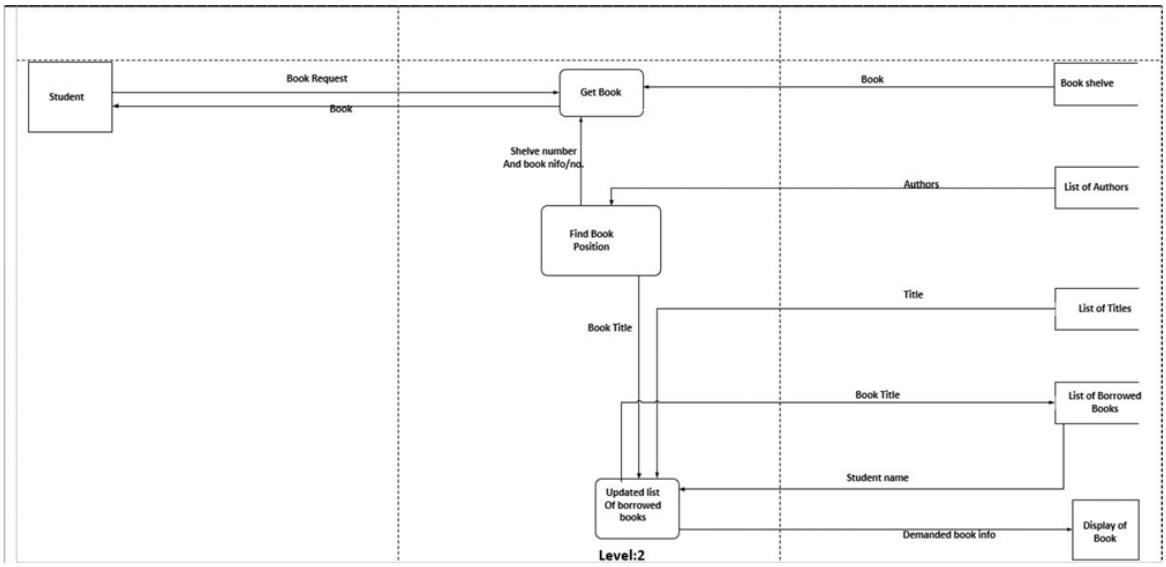
2. DATA FLOW DIAGRAMS



2.1 DFD: LEVEL 0

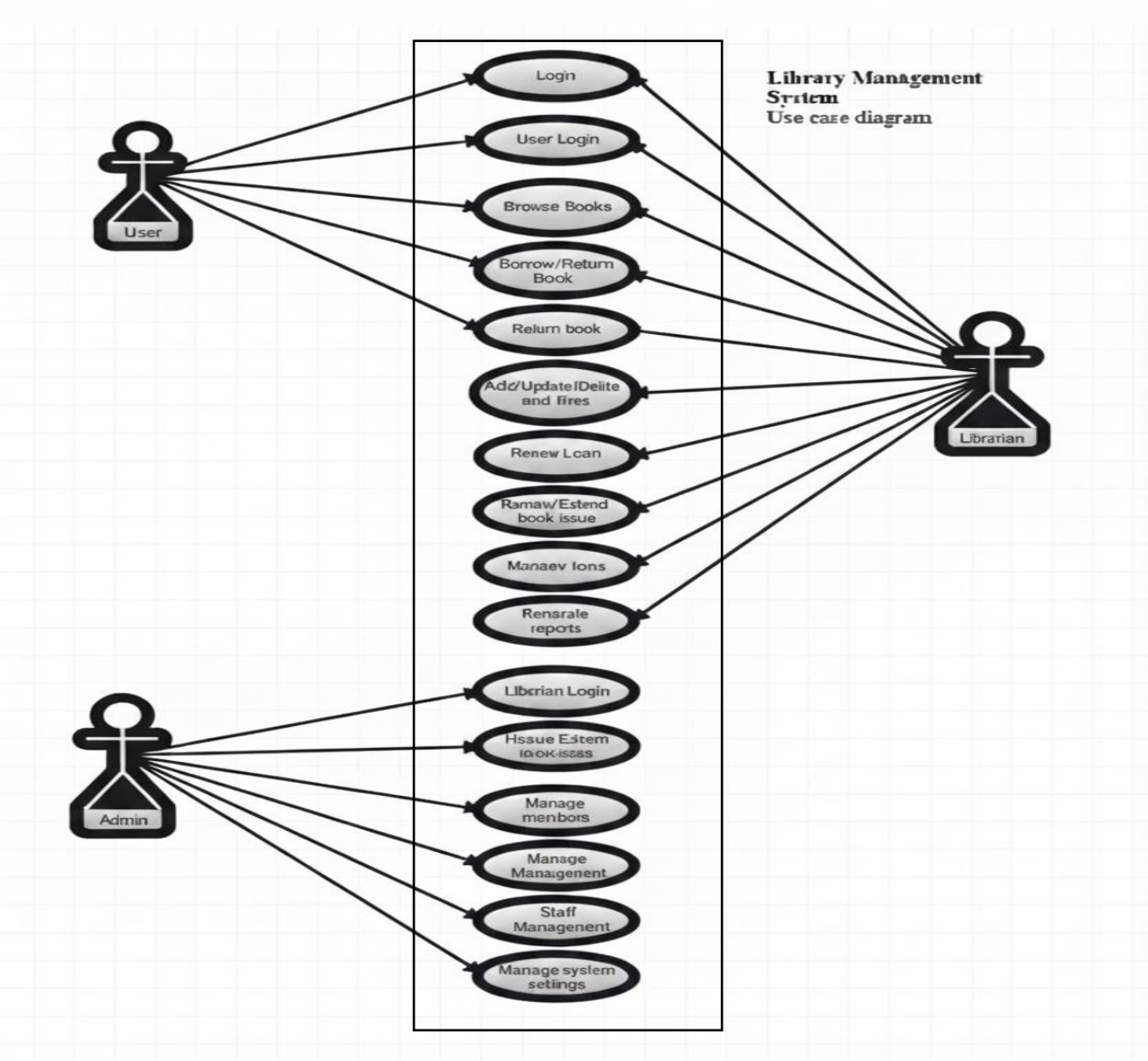


2.2 DFD: LEVEL 1

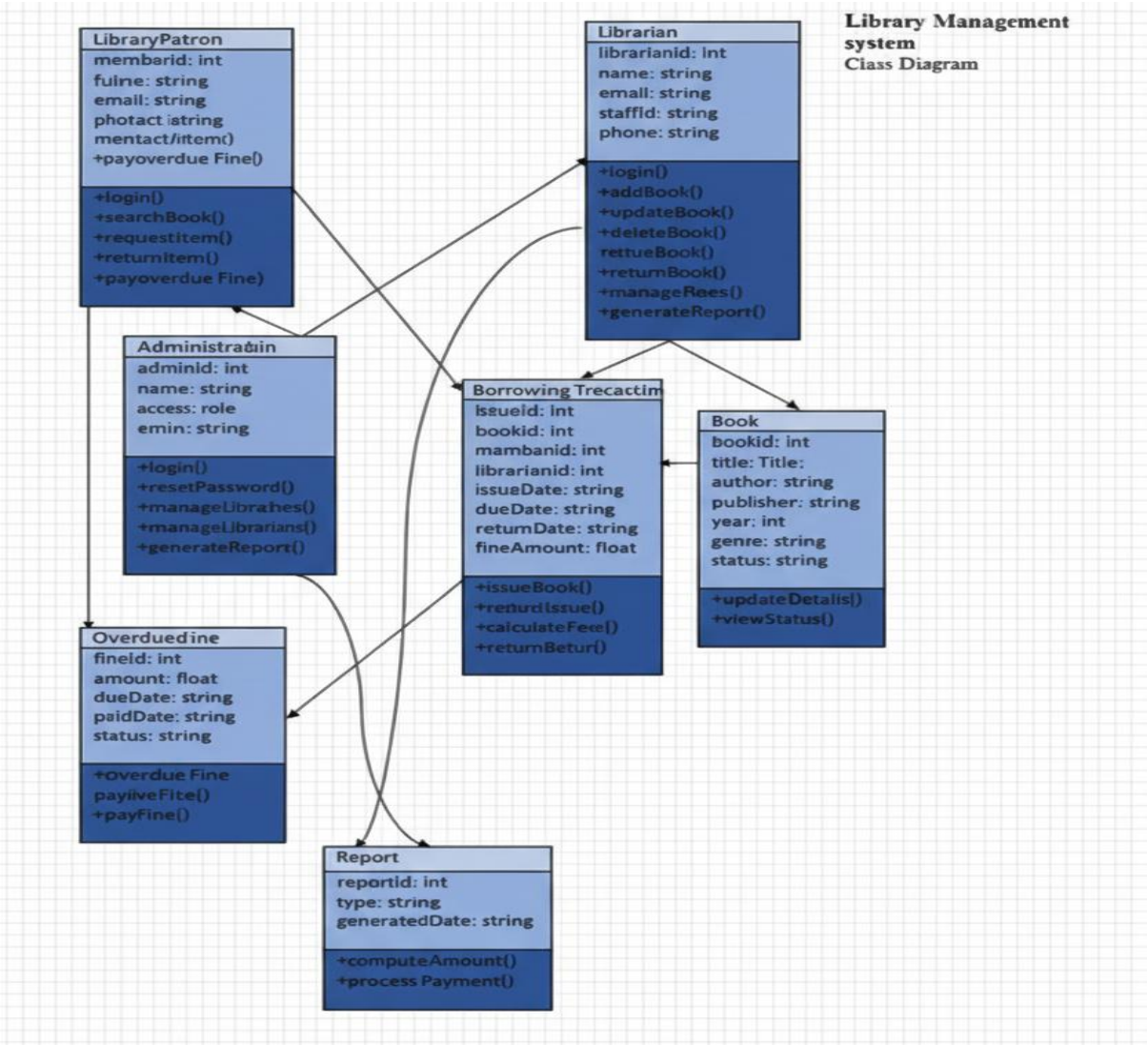


2.3 DFD: LEVEL 2

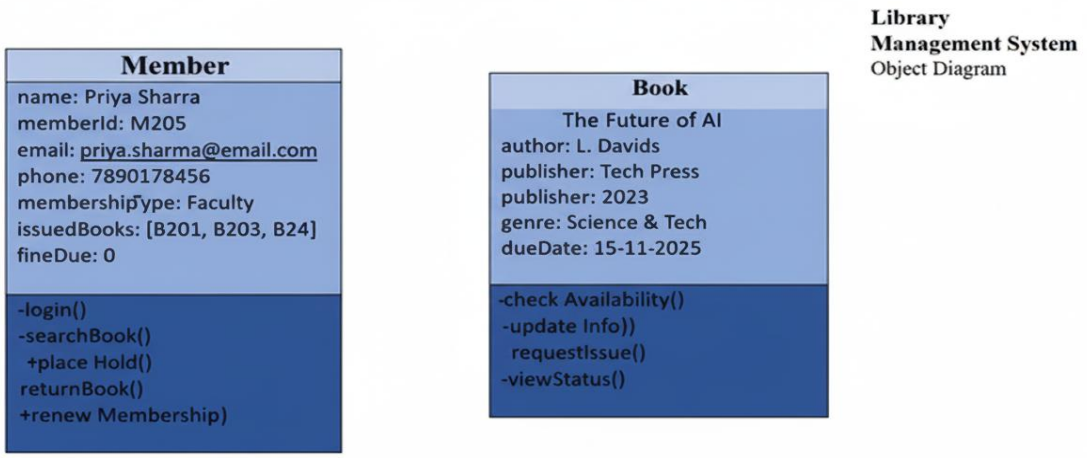
3.UNIFIED MODELING LANGUAGE DIAGRAMS



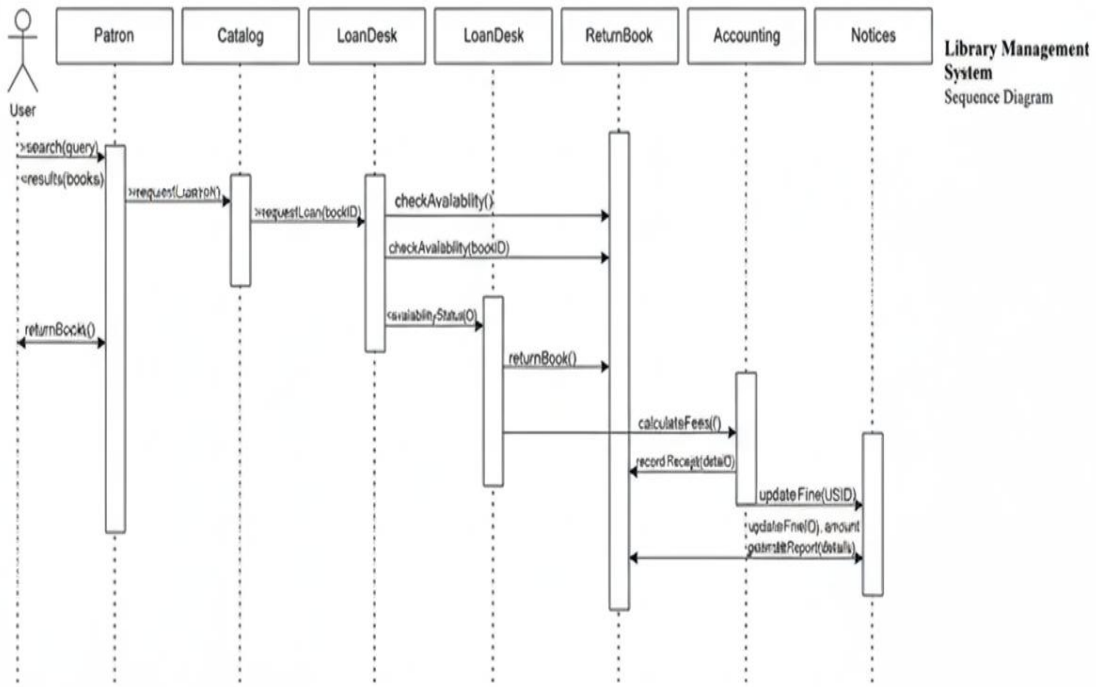
3.1 USE CASE DIAGRAM



3.2 CLASS DIAGRAM



3.3 OBJECT DIAGRAM



3.4 SEQUENCE DIAGRAM

3.5 ACTIVITY DIAGRAM

4. TEST CASE DESIGN

	A	B	C	D	E	F	G	H	I
1	**Test Scenario ID**	**Login-1**	**Test Case ID**	Search-TC-001					
2	**Test Case Description**	**Search Book Functionciality**	**Medium**	High					
3	**Pre-Requisite**	**Search Book Functionciality** diaabase	User is logged in	User is matchig "Invally Fahilur" are displayed					
4	**Test Execution Steps**								
5	**S.No.**	**Action**	**Inputs**	**Expected Output**	**Actual Output**	**Test Browser**	**Test Result**	**Test Comments**	
6	1	Search by Title	Username: lirt the Rings" Password: lib123	Books matching "The Lord of the Rings" are displayed	Displayed 3 results for	Chrome	Pass	Relevaut books found.	
7	2	Search with Invalid Input	"xyx123abc"	No osssage: "No matching books tho books found."	Chrome	Chrome	Pass	Relevem crashed Neeils based tio fix.	

	A	B	C	D	E	F	G	H
8								
10								
11	**Test Scenario ID**	Profile-1	**Test Case ID**	Rofix-01				
12	**Test Case Description**	User must beven Testing	**Tost Priority**	High				
13	**Pre-Requisite**	User must logged in and on profile page.		user can search information is updated.				
14	Test Execution Steps:							
15	**S.No.**	**Action**	**Inputs**	**Expected Output**	**Actual Output**	**Test Browser**	**Test Result**	**Test Comments**
16	1	Update Name & Email Name: Jane Doe	Username: librarian Password: "fython Quiry: "jame smeli sooly jane.doe@email.com	User profin and successfully updated	Profile updated, updated, confirmation message		Pass	[Testn] 15/10/2025 10:30 AM] Login and Search integration verified
10	2	Change Password Current Pass: "oldPass123" Corrrim Pass: "NewPass!23" Iser logged out after after session requetully	Firefox	Password changed, Password changed, log out session terminated	Chrefox	Chrome	Pass	[Testn] 15/10/2025 19:35 AM] Login and Delote integration verified

	A	B	C	D	E	F	O	H	I	J	K
1	**Issue-1**	Issue-1	**Test Case ID**	Search-1A							
2	**Test Case Description**	Search Book - Unit Testing		High							
3	**Pre-Requisite**	User is logged in as Libarian		High	User status updated to displayer	"Issued"	record created				
4	**Test Execution Steps:**										
5	**S.No**	**Action**	**Inputs**	**Expected Output**	**Actual Output**	**Test Browser**	**Test Result**	**Test Comments**			
6	1	Issue Available Book	LibearianID: lib001 MemberID: mem05	Book titled "introductiorly. algorithms" Due. date date calculated.	Book issued Successfulu Due Date: 2025-11-01	Chrome	Pass	[Suraj 14/10/2025 11:55 AM]: Book search succest.			
7	2	Issue Unaailailatleg book	LibearianID: lib001 MemberID: mem05 BookID: B123 (already issued)	Chrome	Message: "Book currently unaailable."	Chrome	Pass	[Suraj 14/10/2025 11:54 AM]:Search working for expected			

	A	B	C	D	E	F	G	H	I	J	K
10			**Test Case ID**	Return-2B							
11	**Loan Management**	**Test Case** **Book Return Integration**	**Test Priority**	High							
12		User has an issued book	User has an issued	User can search his book	Fine is caturbated (any)						
13	Test Execution Steps:										
15	**S.No**	**Action**	**Inputs**	**Expected Output**	**Actual Output**	**Test Browser**	**Test Result**	**Test Comments**			
15	1	Return On-Time Book	Username: member99 Password: ABC123	Book marked as retuned, no fine charged	Book marked as reotuned, no fine fine charged	fine 0	Pass	[User 15/10/2025 09:30 AM] and burn integration verified			
16	2	Return Late Book	Username: member99 Passwords:lb@123 Book ID: XYC456	Book returned as fine calculated	Book retuned, ins successfuled, fine $5.00	Chrome	Pass	[User 15/10/2025 09:30 AM] Search integration verified			
23											

	A	B	C	D	E	F	G	H	I	J	K
22	**Test Scenario ID**	**Fine Management**	**Delete-1A**	**Migh**							
	Test Case	Calculate Fine	Calculate Fine	High							
25	**Pre-Sequlaite**	User is ust bn valid	Hcen	user can delete book							
25	**Tolle**	User inunt be valid	User can doiwn delelee book	Bestirlot Bookis							
27	Test Execution Steps:										
37	**S.No**	**Action**	**Inputs**	**Expected Output**	**Actual Output**	**Test Browser**	**Test Result**	**Test Comments**			
29	1	Waive Fine	Book ID: B005	BooDate: 20.00	Duok ID: B000	Firss	Pass	Pass			
33	2	E Waive Fine	Calculate Fine for Odroue Book Metot: "Ovierdud applied Book	Message "fOverdueate "Tne "Fous" nd i qpthuN"	Fine applied	Fiss	Pass	[Suraj 15/20/2025 Idoleis Forbid Ison Pewtie Fier			
30	3	Fitinuie Fine	Fine ootodi: $3.Δ0, Message Duek :Tib Baiate Fine Mook tnccssfully	"fFleoxe Fine "Fine:t waived	Waived	Pass	Pass	Fiive: Itscbogs oud Fe dvage 25 Firagh iiout Pass			

	A	B	C	D	E	F	G	H	I	J	K
35		**Delete-1A**	**Delete-1B**	**Repat-1B**							
30	1	Lomin + Bopcais	Generate Eoot:d "for ifUpsiroord Book"	Fine sniitdervtxd in conk detkated nord Oct 2021	User depcifn oivrordia iirt odeted successfully	Tast	Pass	[Suraj 15/10/2025 13.45 AM] Search indegotion verfied			
35	2	Generate Report	Geport) Type: "Oatabasn Report" Loui ID: P800	Flonic Ueord ane vinot evnrd dclated Successfully	FHOR ragsk dstated nuccersfully	Chrome	Pass	[Suroj 15/10/2025 13:35 AM] Sesik + Delete Delete Integration verfied			
30	4	Period: Last Miettek	Rocel found ' late Fee Report Report	Generate gererate Gearch : Report Sevarsiomo ee Vsirt	Lirtdl 5 recpod orel generated report dirbelt Oct 2025	Chrome	Pass	[Suraj 16/10/2025 13:09 AM] Cutrctive Reerkrationy Rwies verfied			